GGG-B-001013 (GSA-FSS)
October 19, 1967

INTERIM FEDERAL SPECIFICATION

BIT, SCREWDRIVER (TORQ-SET)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers power driver bits for driving and removing screws having torq-set type recesses.

1.2 Classification. Bits shall be of the following types, as specified (see 6.2).

Type I. Flush use.
Type II. Counterbore use.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

QQ-T-570 - Tool Steel, Alloy.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francsico, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424 - Hand Tools, Packaging of.

Military Standards:

MS 33781 - Recess Torq-Set.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-8C - Dimensioning and Tolerancing.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5130

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3. REQUIREMENTS

3.1 Illustrations. Dimensioning and tolerancing of figures herein shall be interpreted, as specified in MIL-STD-8C. All dimensions are in inches.

3.2 Material. Bits shall be made from high speed tool steel equal in strength and impact resistance to type S2 of QQ-T-570.

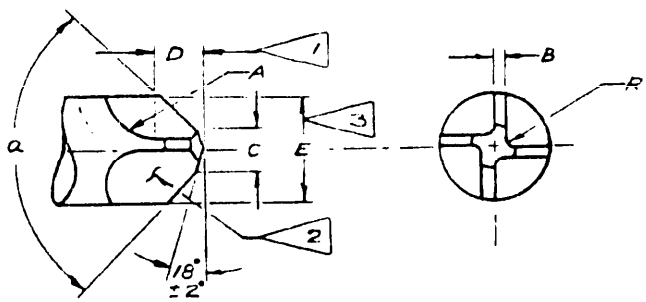
3.3 Hardness. The bits shall be hardened throughout to Rockwell C 56 minimum (see 4.4.1).

3.4 Finish. Finish shall be in accordance with good commercial practice with no presence of burrs or cracks. Surfaces shall be sufficiently smooth to adequately determine dimensions.

3.5 Coating. A rust preventive coating adequate for interim storage shall be applied.

3.6 Marking. Bits shall be marked with the applicable point size (see tables) and with the manufacturer's name or trademark.

3.7 Point design. Bit points shall be as shown in figure 1 and table I for the point size specified (see 6.1).



NOTE: 1. Straight before radius.
2. Slight material build-up permissible beyond area "D".
3. See tables III and IV.
FIGURE 1. Point design.

TABLE I. Point design

Point size	A	B		C		D	R		a
	±1/32	Max.	Min.	Max.	Min.	Min.	Max.	Min.	±0°30'
0	1/8	.0160	.0155	.039	.037	3/64	.005	.000	93°00'
1	1/8	.0160	.0155	.046	.044	3/64	.005	.000	93°00'
2	1/8	.0160	.0155	.054	.052	1/16	.005	.000	93°00'
3	1/8	.0175	.0170	.061	.059	1/16	.005	.000	93°00'
4	1/8	.0195	.0190	.069	.067	5/64	.010	.007	93°00'
5	1/8	.0225	.0220	.077	.075	5/64	.010	.007	93°00'
6	1/8	.0255	.0250	.084	.082	3/32	.016	.013	93°00'
8	1/8	.0305	.0300	.098	.096	3/32	.016	.013	93°00'
10	1/8	.0355	.0350	.113	.111	7/64	.023	.020	93°00'
1/4	3/16	.0475	.0455	.149	.147	1/8	.023	.020	91°00'
5/16	3/16	.0605	.0585	.187	.185	9/64	.028	.023	91°00'
3/8	3/16	.0720	.0700	.223	.221	11/64	.028	.023	91°00'
7/16	1/4	.0850	.0830	.259	.257	11/64	.040	.035	91°00'
1/2	1/4	.0960	.0940	.295	.293	13/64	.055	.050	91°00'
9/16	1/4	.1080	.1055	.331	.329	7/32	.068	.063	91°00'
5/8	1/4	.1200	.1175	.368	.366	1/4	.068	.063	91°00'

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3.8 Shank design. Bit shanks shall be as shown in figure 2 and table II for the shank size specified (see 6.2).

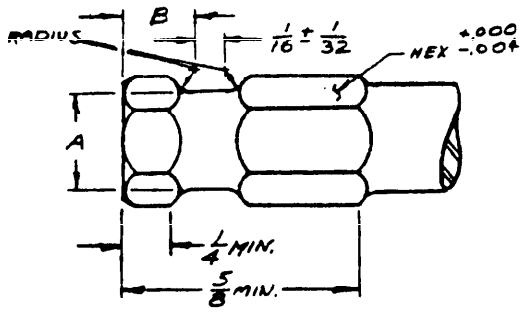


FIGURE 2. Shank design.

TABLE II. Shank design

Shank size (Hex.)	A ±1/32	B ±1/32
1/4	1/8	3/8
5/16	3/16	13/32
7/16	3/16	1/4
5/8	1/4	11/32

3.9 Performance. The bits shall be capable of meeting the torque test of 4.4.2.

3.10 Type I. Type I bits shall be suitable for normal (flush) driving applications and conform to figure 3 and table III for the point size and shank size specified (see 6.1).

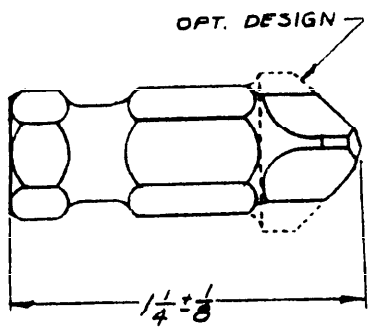


FIGURE 3. Type I, flush use.

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TABLE III. Type I, flush use.

Point size	Hex size	E (fig. 1) max.	E (fig. 1) min.	Point size	Hex size	E (fig. 1) max.	E (fig. 1) min.
0	1/4	.185	.180	1/4	5/16	.437	.429
1	1/4	.185	.180	1/4	7/16	.434	.429
2	1/4	.185	.180	5/16	1/4	.437	.429
3	1/4	.185	.180	5/16	5/16	.437	.429
4	1/4	.185	.180	5/16	7/16	.434	.429
5	1/4	.185	.180	3/8	1/4	.504	.496
6	1/4	.185	.180	3/8	5/16	.504	.496
8	1/4	.247	.242	3/8	7/16	.423	.429
8	5/16	.309	.304	7/16	7/16	.423	.429
10	1/4	.247	.242	1/2	5/8	.622	.617
10	5/16	.309	.304	9/16	5/8	.622	.617
1/4	1/4	.437	.429	5/8	5/8	.622	.617

3.11 Type II. Type II bits shall be suitable for use in counterbored holes and conform to figure 4 and table IV for the point size, and length specified (see 6.1).

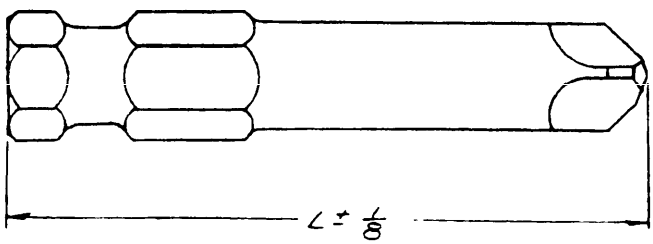


FIGURE 4. Type II, counterbore use.

TABLE IV. Type II, counterbore use

Point size	$E^{+0}_{-1/64}$ (fig. 1)	Shank size	L			
0	1/8	1/4	1-7/8	2-3/4	3-1/2	6
1	1/8	1/4	1-7/8	2-3/4	3-1/2	6
2	1/8	1/4	1-7/8	2-3/4	3-1/2	6
3	5/32	1/4	1-7/8	2-3/4	3-1/2	6
4	5/32	1/4	1-7/8	2-3/4	3-1/2	6
5	3/16	1/4	-----	-----	3-1/2	6
6	3/16	1/4	1-7/8	2-3/4	3-1/2	6
8	1/4	1/4	1-7/8	2-3/4	3-1/2	6
10	1/4	1/4	1-7/8	2-3/4	3-1/2	6
1/4	11/32	7/16	-----	-----	3-1/2	6
5/16	3/8	7/16	-----	-----	3-1/2	6
3/8	7/16	7/16	-----	-----	3-1/2	6
7/16	7/16	7/16	-----	-----	3-1/2	6

3.12 Workmanship. Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table V.

TABLE V. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 ea.	II	1.5	Percent defective	4.3.1
Dimensional examination	1 ea.	II	1.5	Percent defective	4.3.2
Testing	1 ea.	S-4	1.5	Percent defective	4.4
Preparation for delivery examination	one container	S-2	4	Defects per hundred units	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in finish, coating, marking, design, or workmanship.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing.

4.4.1 Hardness test. Hardness tests shall be conducted in accordance with Federal Standard No. 151 to determine compliance with 3.3.

4.4.2 Torque test. The bit point shall be inserted in a screw or test block having a recess in accordance with MS 33781 (flat fillister head). The required torque loads shown in table VI shall be applied to the shank while restraining the screw. There shall be no evidence of distortion or breakage at the conclusion of the test.

TABLE VI. Torque loads

Point size	Load (inch-pounds)	Point size	Load (inch-pounds)
0	*	10	100
1	*	1/4	250
2	10	5/16	500
3	15	3/8	900
4	25	7/16	1300
5	35	1/2	2200
6	45	9/16	2400
8	75	5/8	3000

* No test required.

4.5 Inspection of preparation for delivery. Inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with Military Specification MIL-H-15424.

5. PREPARATION FOR DELIVERY

(Nonmilitary agencies. Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging.

5.1.1 Level A. Cleaning, preservation, and packaging shall be in accordance with level A of Military Specification MIL-H-15424.

5.1.2 Level C. Preservation and packaging which will afford adequate protection against corrosion, deterioration, and damage during shipment from supply source to the first receiving activity for immediate use: (vendor to user).

5.2 Packing.

5.2.1 Level A. Packing shall be in accordance with level A of Military Specification MIL-H-15424.

5.2.2 Level B. Packing shall be in accordance with level B of Military Specification MIL-H-15424.

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5.2.3 Level C. Packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.3 Marking.

5.3.1 Military agencies. In addition to any special marking required by the contract, or order, or herein, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. In addition to any special marking specified in the contract or order (see 6.1), each unit and intermediate package and shipping container shall be marked in accordance with Federal Standard No. 123.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurements documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Size required, state point size and shank size on type I, and point size and length on type II (see table VII).
(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) Marking, if different (see 5.3).

6.2 Availability. Commercially standard bits are as shown in table VII.

TABLE VII. Availability

Type I (All are 1-1/4 nom. length)					Type II				
Point size	Shank size				Shank size	Nominal length			
	1/4	5/16	7/16	5/8		1-7/8	2-3/4	3-1/2	6
0	A*				1/4	A	A	A	A
1	A				1/4	A	A	A	A
2	A				1/4	A	A	A	A
3	A				1/4	A	A	A	A
4	A				1/4	A	A	A	A
5	A				1/4	A	A	A	A
6	A				1/4	A	A	A	A
8	A	A			1/4	A	A	A	A
10	A	A			1/4	A	A	A	A
1/4	A	A	A		7/16			A	A
5/16	A	A	A		7/16			A	A
3/8	A	A	A		7/16			A	A
7/16			A		7/16			A	A
1/2				A					
9/16				A					
5/8				A					

*A - Available

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